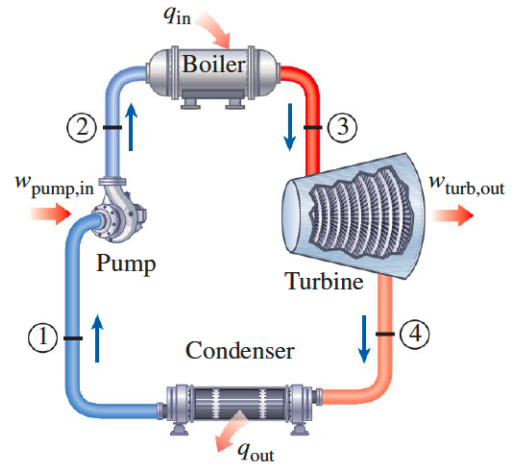


Chapter 8 Workshop

1. A simple ideal Rankine cycle which uses water as the working fluid operates its condenser at 40°C and its boiler at 250°C . Calculate the work produced by the turbine, the heat supplied in the boiler, and the thermal efficiency of this cycle when the steam enters the turbine without any superheating



2. A refrigerator uses refrigerant-134a as the working fluid and operates on the ideal vapor-compression refrigeration cycle. The refrigerant enters the evaporator at 120 kPa and leaves the compressor at 1 MPa. If the compressor consumes 450 W of power, determine (a) the mass flow rate of the refrigerant and (b) the COP of the refrigerator.

